A REPORT

ON

AI-ASSISTED DREDGING ASSESSMENT FOR INDIAN PORTS: IMPLEMENTATION OF THE HAINE HYBRID NUMERICAL–MACHINE LEARNING FRAMEWORK (KOCHI PILOT)

BY

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Prepared in partial fulfillment of the
Practice School – I

AT

Maritime Research Centre (MRC) – Foundation for Underwater Domain Awareness

A Practice School – I

Station of

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

June 2026

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

Practice School Division

Station: Maritime Research Centre (MRC) **Centre:** Pune, Maharashtra

Duration: 7 weeks **Date of start:** 12 May 2026

Date of submission: 18 June 2026

Title of the project: AI-Assisted Dredging Assessment for Indian Ports — Implementation of the HAINE Hybrid Numerical–Machine Learning Framework (Kochi Pilot)

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Key words: Dredging; siltation prediction; TELEMAC-2D; satellite SPM; cohesive sediment; nautical depth; under-keel clearance; Cochin Port; decision support

Project areas: Coastal hydrodynamics, remote sensing, sediment transport, applied machine learning, maritime infrastructure

Abstract: This report documents the implementation of the first working pilot of HAINE (Hybrid AI–Numerical Engine), a four-layer framework for AI-assisted dredging assessment, applied to Cochin (Kochi) Port. A validated TELEMAC-2D tidal hydrodynamic model (Layer 1) was coupled to a gap-free thirteen-month satellite suspended-particulate-matter cube built from Sentinel-2 and Landsat imagery (Layer 2). A physics-based coupling gates a calibrated cohesive-deposition sub-model that reproduces the documented order-of-magnitude annual siltation, disaggregated by navigation zone against published fluid-mud structure; an under-keel-clearance calculator and a decision dashboard complete the chain (Layers 3–4). The pilot is delivered honestly as a literature-calibrated estimate rather than a survey-trained model, and the report catalogues the engineering corrections required, quantifies where the predictions are trustworthy and where they deviate, and specifies the data and modelling upgrades needed to close the remaining gaps.

Signature of the student Signature of the PS faculty member

Date Date

Acknowledgements

I express my sincere gratitude to the Maritime Research Centre (MRC) and the Foundation for Underwater Domain Awareness (FUDA), Pune, for hosting this Practice School and for framing the problem within the wider MAHASAGAR maritime vision. I thank my mentors, Jay Pinjarkar and Shridhar Prabhuraman, for their sustained technical guidance and for encouraging a physics-first, honestly-scoped approach to the work.

I acknowledge the open scientific infrastructure without which this implementation would not have been possible: the openTELEMAC community for the hydrodynamic solver, the European Space Agency Copernicus programme and the USGS/NASA Landsat programme for freely available imagery through Google Earth Engine, and ECMWF for the ERA5 reanalysis. The published work of the Cochin siltation community, in particular the nautical-depth study of Ferket and colleagues, provided the physical anchors used to calibrate the pilot.

Finally, I thank the Practice School Division of BITS Pilani and my family for their support throughout the internship.

Signature of the student

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Chapter 1

Introduction

India operates twelve major ports and more than two hundred non-major ports, and almost every one of them shares a single chronic operational burden: sedimentation of approach channels, turning circles and berth pockets. Cochin Port alone accounts for the largest annual maintenance-dredging quantity of any Indian major port, of the order of twenty-four million cubic metres per year. Dredging is currently procured reactively — a bathymetric survey reveals a shoal, a work order is raised, and a contractor is mobilised — with no predictive, physically-grounded intelligence about *where*, *when* and *how much* siltation will occur. This report describes the implementation of a system designed to supply exactly that intelligence for the tropical estuarine ports of the Indian Ocean Region, demonstrated on Cochin as a representative high-siltation pilot.

1.1 Motivation and the Tropical Modelling Gap

Established sediment-transport software is calibrated for temperate estuaries. Indian ports differ in three ways that a temperate model does not capture: a monsoon that concentrates the majority of the annual sediment supply into roughly four months and simultaneously blacks out optical satellites with cloud; warm ($\sim 30^{\circ}$C) waters that lower the critical shear stress of cohesive mud and mobilise sediment that would otherwise be stable; and heavily dammed rivers whose reduced fluvial supply shifts the siltation balance towards locally re-suspended, coastally-derived material. A useful tool for Indian ports must therefore fuse a hydrodynamic engine with satellite observation and with a calibration strategy that respects these tropical realities. HAINE was conceived as that fusion, and this report reports its first end-to-end implementation.

1.2 Objectives of the Implementation

The Practice School objectives were deliberately implementation-level rather than purely conceptual:

1. to stand up a working hydrodynamic model of the Cochin estuary and obtain a validated tidal solution;
2. to build a gap-free, physically-validated satellite suspended-sediment product over the port area of interest;

3. to couple the two so that hydrodynamics genuinely gates a sediment-deposition estimate, and to produce a first, calibrated siltation-volume assessment together with an under-keel-clearance calculator and a decision dashboard;

4. to do all of this *honestly* — documenting every approximation, quantifying where the outputs can be trusted, and specifying precisely what additional data would convert the pilot from a calibrated estimate into a trained predictive model.

1.3 Scope and Organisation

The pilot is a proof of concept, not an operational product. Layer 1 is implemented as a tidal-only hydrodynamic solve; wave forcing and a solver-coupled river run are identified as upgrades rather than delivered. Layer 3 is delivered as a calibrated physical estimate because the ground-truth labels required to train a machine-learning model are institutionally gated. Chapter 2 reviews the relevant literature. Chapter 3 describes the framework as actually implemented. Chapter 4 presents the results. Chapter 5 catalogues the engineering problems encountered and the corrections applied. Chapter 6 quantifies prediction validity. Chapters 7 and 8 draw conclusions and make recommendations.

Chapter 2

Literature Review

The implementation draws on four strands of literature: numerical hydrodynamics and cohesive-sediment transport, satellite retrieval of suspended sediment and bathymetry, machine-learning approaches to sedimentation, and the specific body of work on Cochin.

2.1 Hydrodynamic and Cohesive-Sediment Modelling

Free-surface hydrodynamics on unstructured meshes is a mature field; the finite-element formulation underlying the TELEMAC system [4] is well suited to the complex coastline of an estuarine port and was adopted here in preference to a structured-grid approach. Cohesive erosion and deposition are universally described by the Partheniades–Krone framework [6], in which deposition proceeds only where the bed shear stress falls below a critical value; this thresholding is the physical mechanism that HAINE exploits to let hydrodynamics gate deposition. Regional applications of coupled wave–flow models on the Indian west coast [7] confirm that littoral and monsoonal forcing dominate the sediment budget, motivating the identification of wave coupling as a priority upgrade.

2.2 Satellite Retrieval of Suspended Sediment and Depth

The single-band turbidity algorithm of Nechad and colleagues [10] provides a generic, multi-sensor retrieval of total suspended matter that transfers across Sentinel-2 and Landsat red bands, and it forms the basis of the Layer 2 product. Where multibeam bathymetry is unavailable, the blue/green log-ratio method of Stumpf and colleagues [17] yields satellite-derived bathymetry to metre-scale accuracy in clear water, though its reliability degrades in the high-turbidity monsoon conditions typical of Indian approaches. The estuarine turbidity maximum of the central west coast [5] provides the observational context against which the fused product is validated.

2.3 Machine Learning for Sedimentation

Recent globally-scalable frameworks such as RECLAIM [9] demonstrate that gradient-boosted models can predict reservoir and channel sedimentation where sufficient labelled

survey data exist. The central lesson taken from this literature, reinforced by critical reviews of machine learning in ocean remote sensing [11], is that such models are only as good as their labels; in the absence of paired survey data a calibrated physical model is the honest alternative, and is the path taken here.

2.4 Cochin-Specific Studies and Environmental Context

The definitive contemporary study of Cochin siltation [3] reports the largest siltation rate among Indian ports, quantifies per-zone fluid-mud thicknesses, and establishes the nautical-depth density criterion of approximately 1170 $\mathrm{kg\,m^{-3}}$ used in this work; critically, it coupled a wave model, underlining the limitation of a tidal-only Layer 1. Measured sediment densities for the Cochin backwaters [14], the estuary's tidal constants [1] and its river discharge regime [16] supply the physical constants used in the pilot. The ecological cost of dredging is documented for the same estuary [15] and more broadly for Indian coastal ecosystems [13], motivating the environmental-risk framing of the decision layer. Strategic direction is provided by the MoPSW Dredging Guidelines [8] and the MAHASAGAR vision [12], while tidal-inlet performance scoring [18] and the damming–sedimentation link [19] inform the Kochi-specific adaptations.

Chapter 3

Methodology: The HAINE Framework as Implemented

HAINE is organised as four coupled layers, shown in Figure 3.1. Layer 1 supplies the physics, Layer 2 supplies observation, Layer 3 fuses them into decision-relevant sub-models, and Layer 4 presents the result. This section describes what was actually built, including the substitutions made relative to the original design.

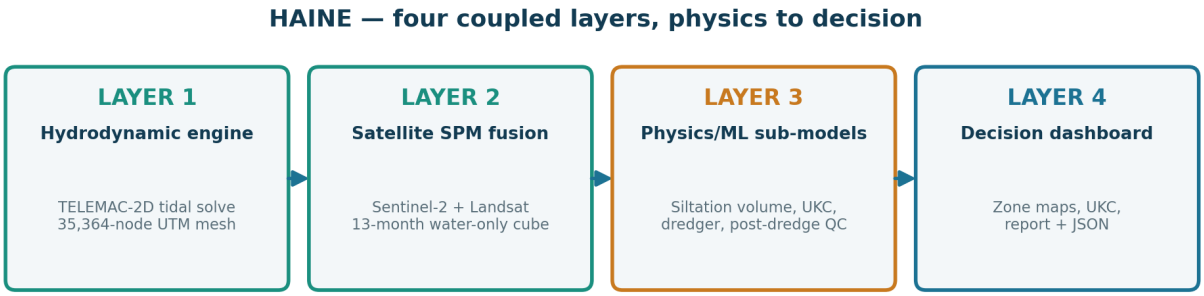


Figure 3.1: The four-layer HAINE architecture as implemented for the Kochi pilot, with the principal tools and outputs of each layer.

3.1 Layer 1: Hydrodynamic Engine

The original design specified Delft3D-FM. During implementation the engine was changed to openTELEMAC (version 8p5r0), an open-source finite-element solver, because it proved more reproducible and scriptable within a Linux (WSL2) environment and better suited to the unstructured triangular mesh. The shallow-water equations are solved on a mesh of 35,364 nodes and 69,598 triangular elements covering the Kochi estuary and adjoining coast; the bed elevation, shown in Figure 3.2, is interpolated from a fused satellite-derived and regional bathymetry grid.

Cohesive deposition follows the Krone term

$$D = w_s\, C \left(1 - \frac{\tau_b}{\tau_{cd}}\right), \qquad \tau_b < \tau_{cd}, \tag{3.1}$$

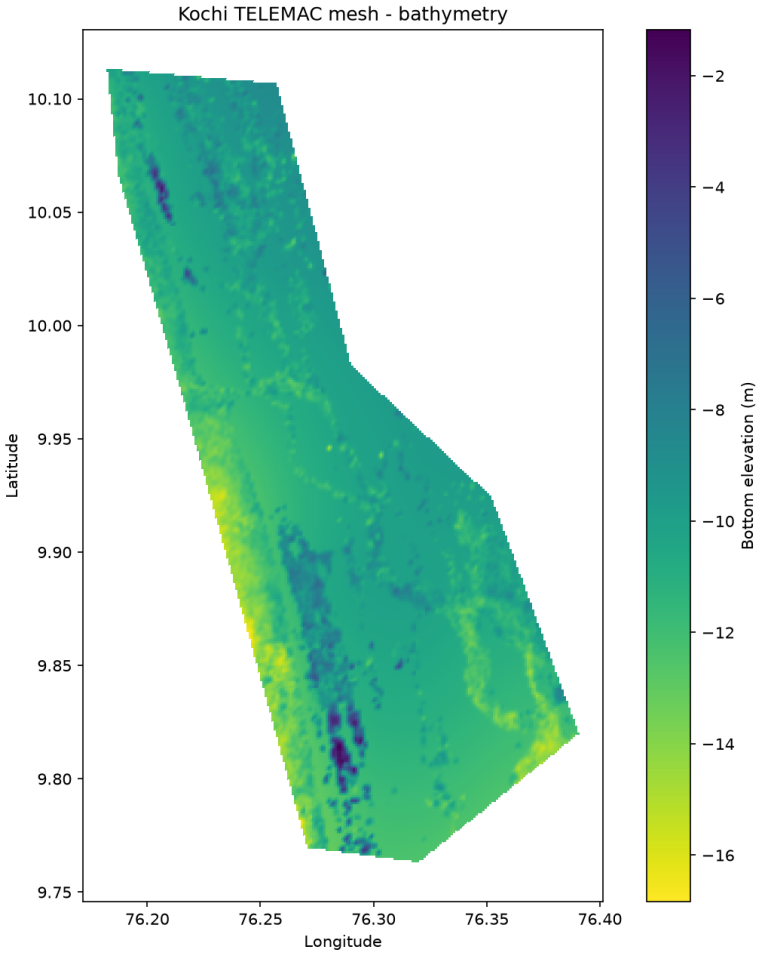


Figure 3.2: The Kochi TELEMAC mesh coloured by bed elevation. The deep navigation channels (light) thread through the estuary; the seaward boundary lies to the west.

where w_s is the settling velocity, C the near-bed concentration, τ_b the bed shear stress and τ_{cd} the critical shear for deposition. Deposition-prone cells are those where $\tau_b < \tau_{cd}$ across the tidal cycle, and it is this thresholding that couples Layer 1 to Layer 3.

Two implementation details proved decisive. First, the mesh must be expressed in a metric coordinate system: solving in geographic degrees rendered the Courant condition meaningless and made every timestep unstable, whereas reprojection to UTM Zone 43N (EPSG:32643) resolved the instability entirely. Second, the boundary must be ordered correctly. The boundary loop, recovered by edge-counting and visualised in Figure 3.3, is a single continuous ring of 1,128 points; the seaward arc of 491 points is set as an open, prescribed-elevation boundary forced by the Cochin tidal constituents, and the remaining estuary edges are walls. The pilot is a tidal-only solve; wave forcing and real tidal phases (from the FES2014 atlas) are identified as upgrades.

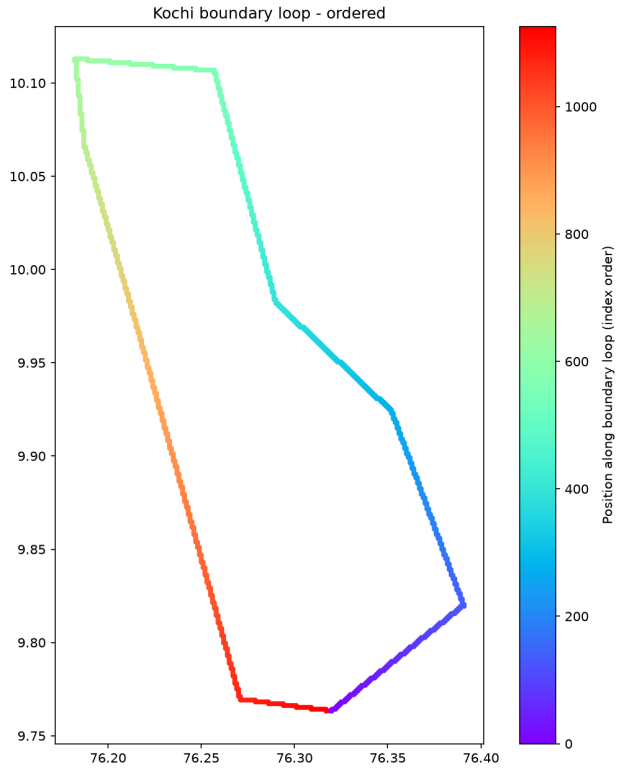


Figure 3.3: The mesh boundary loop coloured by position along the ring (0 to 1,128), confirming a single continuous ordering used to assign open-sea and wall boundary conditions.

3.2 Layer 2: Satellite SPM Fusion

Suspended-particulate-matter fields are retrieved with the Nechad single-band algorithm applied to Sentinel-2 and Landsat-9 red-band reflectance through Google Earth Engine, yielding 223 per-scene rasters. These are grouped by sensor and date and mosaicked — same-date multi-tile exports being genuinely different tiles rather than duplicates — reducing 223 files to 82 unique sensor-dates. Every raster is reprojected onto a common grid in the Layer 1 CRS, and a single global temporal variogram is fitted once and reused per pixel to keep the kriging fusion tractable, producing thirteen monthly fields. These are clipped to the port area of interest, negative undershoots are clamped, and land pixels are removed by a temporal masking rule. The final water-only product is a cube of shape $13 \times 112 \times 133$ in EPSG:32643.

For multi-year operation the extraction is extended backwards in time; because Landsat-9 did not exist before 2021, Landsat-8 is substituted for historical years, and the sparser 2016 record (Sentinel-2A only) is gap-filled by the temporal kriging.

3.3 Layer 3: Coupling and Decision Sub-models

The genuine coupling extracts the depth-averaged velocity from the TELEMAC result at every mesh node and every timestep, forms the tidally-averaged Krone deposition probability

$$\overline{p}_{dep} = \left\langle \max\!\left(0,\, 1 - \tfrac{\tau_b(t)}{\tau_{cd}}\right)\right\rangle_t, \tag{3.2}$$

and interpolates this per-node field onto the Layer 2 SPM grid. Because slack water contributes $p_{dep} \to 1$ at every node, the cycle-averaged field is positive everywhere and varies spatially, which is the physically correct behaviour.

Sub-model A (siltation volume and depth reduction). The monthly SPM field, scaled to a near-bed concentration, is multiplied by the settling flux and the deposition gating and by a parametric Periyar-discharge scour term, then integrated over the maintained-dredging footprint to an annual volume. A single calibration factor scales the estimate to the publicly-reported annual dredging quantity; the total is therefore calibrated, not independently predicted. The spatial distribution across the approach, Ernakulam, Mattancherry and basin zones is anchored to the published fluid-mud thicknesses and textured within each zone by the satellite signal.

Under-keel-clearance calculator. For each channel the available clearance is computed from the charted depth, the spring tidal rise reconstructed from the harmonic constituents, a Barrass squat allowance [2], and a nautical-depth bonus reflecting controlled penetration of the fluid-mud layer to the 1170 $\mathrm{kg\,m^{-3}}$ isopycnal.

Dredger selection and post-dredge quality are rule-based modules keyed to the estimated volume, the soft cohesive sediment type and the nautical-depth concept.

It must be stated plainly that Sub-model A is a *calibrated physical estimate*, not a trained machine-learning model. Two data gaps prevent training: the absence of paired pre- and post-dredge survey rasters (the true per-location labels) and the availability of only a single year of fused SPM at the time of implementation. The framework is deliberately structured so that when survey labels arrive the calibration block is replaced by a trained regressor without altering the rest of the pipeline.

3.4 Layer 4 and Kochi-Specific Adaptations

Layer 4 renders the accretion map, the per-zone volumes, the monthly siltation against river discharge, the under-keel clearances and a sensitivity analysis into a single decision dashboard, exported additionally as a machine-readable results file and an HTML report. Two Kochi-specific physical adaptations are incorporated: an effective-discharge term that separates the fluvially-delivered siltation of the dammed Periyar from the locally-resuspended coastal component, and a tidal-inlet performance index, after Zarzuelo and colleagues [18], that scores candidate dredging strategies for the Cochin Gut against navigation gain, flushing gain and ecological disturbance.

Chapter 4

Results and Discussion

This chapter reports the delivered results layer by layer, and discusses what each does and does not establish.

4.1 Layer 1: Validated Tidal Solution

The tidal simulation completes cleanly over a full 24-hour cycle. The free surface at the seaward boundary oscillates between -0.38 and $+0.63$ m, and the tide propagates into the estuary with physically correct damping and phase lag. The result confirms that the reprojection to a metric CRS produced a stable, well-posed model; the mesh and its bathymetry (Figure 3.2) and the ordered boundary (Figure 3.3) are the validated foundation on which the coupling is built.

4.2 Layer 2: The Fused SPM Cube and Its Validation

The thirteen monthly water-only SPM fields are shown in Figure 4.1. The decisive validation is temporal: the monthly median SPM, Figure 4.2, traces a winter trough near 22 $\mathrm{mg\,L^{-1}}$ rising to a spring peak near 60 $\mathrm{mg\,L^{-1}}$, independently reproducing the documented Cochin monsoon siltation cycle from satellite data alone. This agreement, obtained without any tuning to the siltation record, is strong evidence that the fusion pipeline is physically sound.

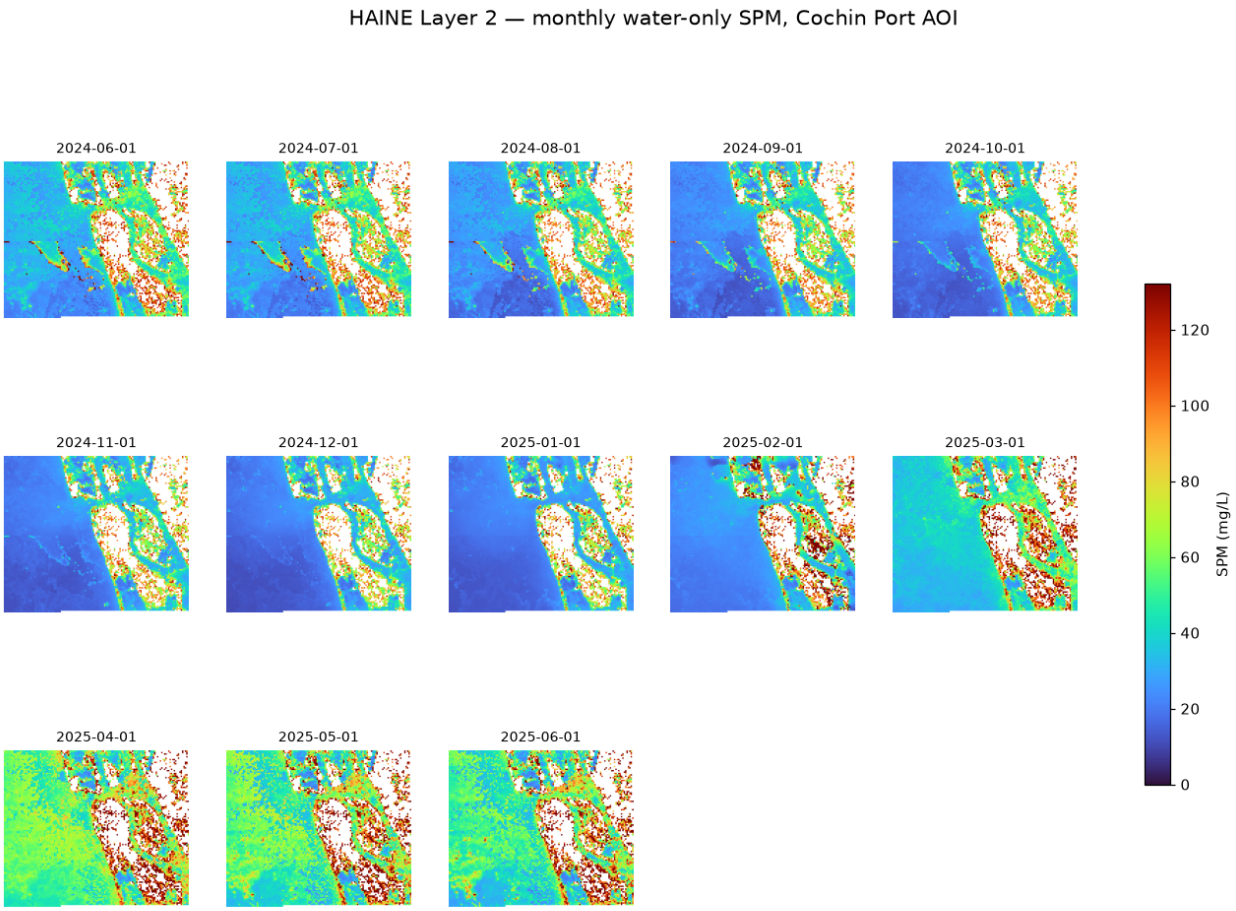


Figure 4.1: The Layer 2 product: thirteen monthly water-only SPM fields over the Cochin Port area of interest, June 2024 to June 2025.

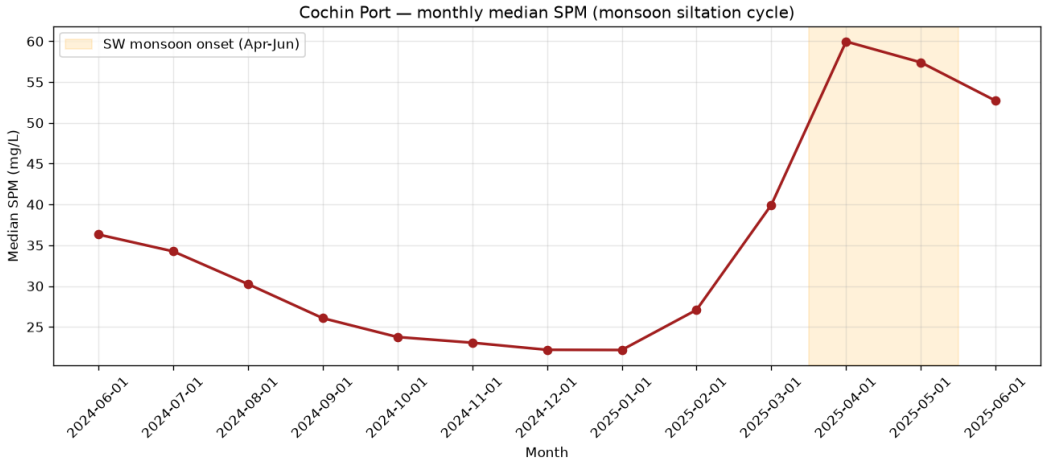


Figure 4.2: Monthly median SPM over the port AOI, reproducing the documented monsoon-driven siltation cycle (winter trough to spring peak).

4.3 Layer 3 Coupling: Deposition Gating and Bed Shear

The coupled fields are shown in Figure 4.3. The tidally-averaged deposition probability (left) is close to unity across most of the port, which is itself an informative result: the port interior is a genuinely low-energy depositional environment in which tidal shear rarely challenges the deposition threshold. The mean bed-shear field (right) nonetheless shows clear spatial structure — a brighter high-shear ribbon along the gut and approach channel against the darker quiescent basins — which is exactly the pattern expected physically. The practical implication, discussed further in Chapter 6, is that under tidal-only forcing the hydrodynamic term is a gentle lever and the siltation estimate is supply-dominated.

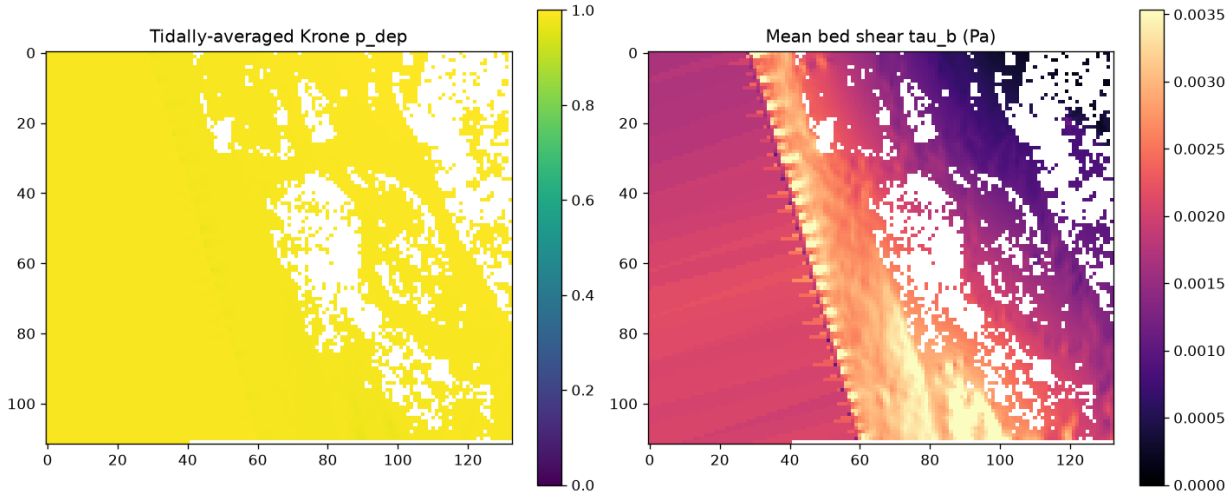


Figure 4.3: Layer 1 to Layer 3 coupling output: tidally-averaged Krone deposition probability (left) and mean bed shear stress in Pa (right) on the SPM grid.

4.4 Sub-model A: Calibrated Siltation Volume

The uncalibrated physical estimate integrated over the whole area of interest substantially overshoots the reported dredging record; restricting the integration to the maintained-dredging footprint reduces it to about 68% of the gross, and a residual factor then scales it to the reported annual quantity. This decomposition, Figure 4.4, replaces an opaque single fudge-factor with an interpretable chain: gross deposition, dredged footprint, residual net-to-gross factor, and the record-anchored total, with the non-dredged estuary siltation reported separately. The per-zone volumes, anchored to the published fluid-mud structure and compared in Figure 4.4, place the largest burden on the outer approach channel, consistent with the literature.

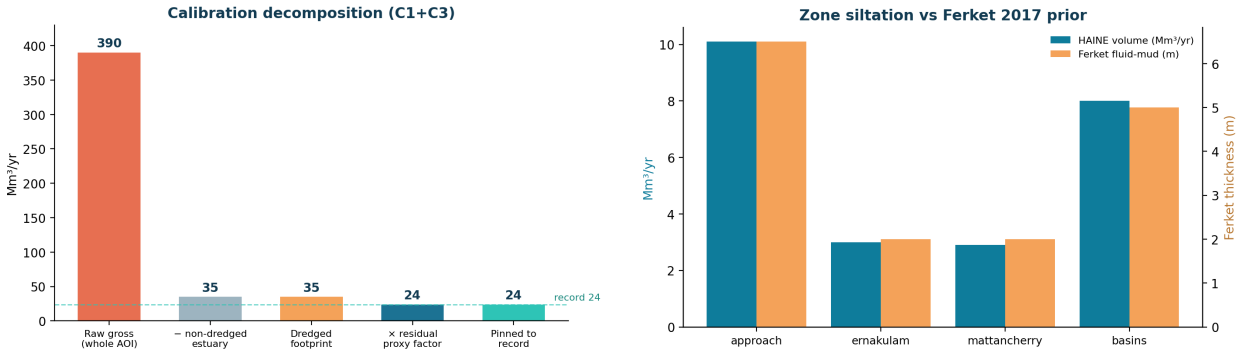


Figure 4.4: Left: decomposition of the calibrated siltation volume from raw gross deposition to the record-anchored total. Right: per-zone siltation volume against the published fluid-mud thickness prior.

4.5 Under-Keel Clearance

The under-keel-clearance results, Table 4.1, are independent of the siltation calibration and are directly usable for tidal-window planning. At spring high tide the approach channel requires an additional 1.6 m of tide to accommodate a 14.5 m-draft vessel, the Ernakulam channel requires 0.95 m, and the Mattancherry channel clears.

Channel	Charted depth (m)	Design draft (m)	Squat (m)	Gross UKC (m)	Status
Approach	14.5	14.5	0.65	0.49	needs +1.6 m tide
Ernakulam	13.25	12.5	0.65	0.94	needs +0.95 m tide
Mattancherry	9.75	9.0	0.65	0.94	clears

Table 4.1: Under-keel clearance at spring high tide (Barrass squat plus nautical-depth allowance).

4.6 Layer 4: The Decision Dashboard

The corrected decision dashboard is shown in Figure 4.5. It integrates the calibrated accretion map, the Ferket-anchored zone volumes and their thickness prior, the under-keel clearances, and a sensitivity analysis showing that the raw volume responds almost linearly to the sediment proxy parameters — a ±20% change in settling velocity, dry density or the near-bed enhancement produces a ±40% change in volume, quantifying the value of measured sediment parameters.

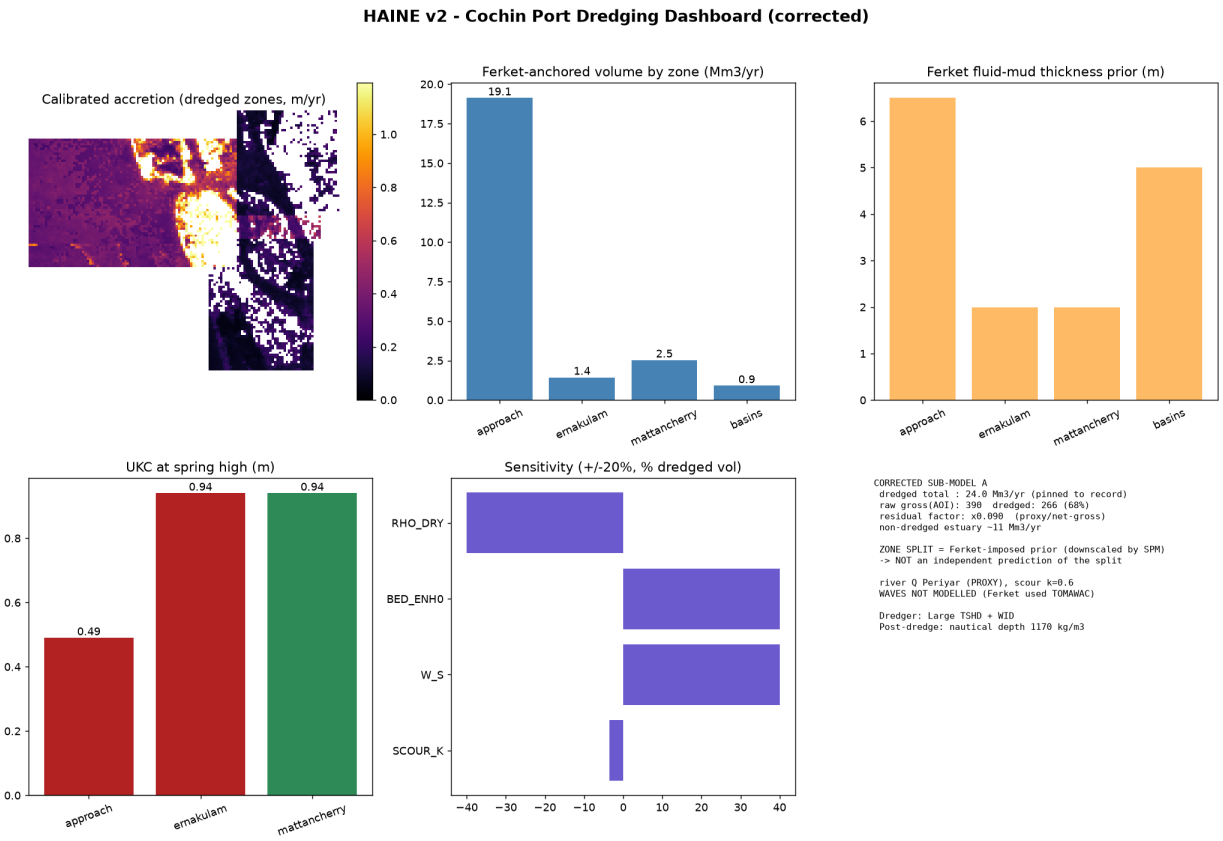


Figure 4.5: The Layer 4 decision dashboard (corrected version): accretion map, zone volumes, thickness prior, under-keel clearance, sensitivity analysis and summary.

Chapter 5

Problems Encountered and Corrections

A substantial part of the implementation effort was diagnostic. This chapter records the principal problems and their corrections, both because they document the work honestly and because they encode hard-won lessons for future sessions.

5.1 Hydrodynamic and Coupling Problems

The most impactful single issue was numerical instability caused by the mesh being expressed in geographic degrees; reprojection to UTM metres resolved every timestep's instability at once. At the coupling seam, two failures appeared in sequence, illustrated conceptually in Figure 5.1. First, a uniform placeholder bed shear exceeded the critical value everywhere and produced zero deposition — a "no-deposition wall" — which was removed by extracting the true per-pixel, tidally-averaged deposition probability from the velocity field. Second, the resulting field was flat because the solver mesh is stored in a local, origin-shifted frame while the SPM grid is in true UTM; the two did not overlap, and every port pixel was filled from a single distant stagnant node. Recovering the origin offset from the geographic mesh (whose node ordering was verified identical) and adding it back co-located the two grids and restored correct spatial gating.

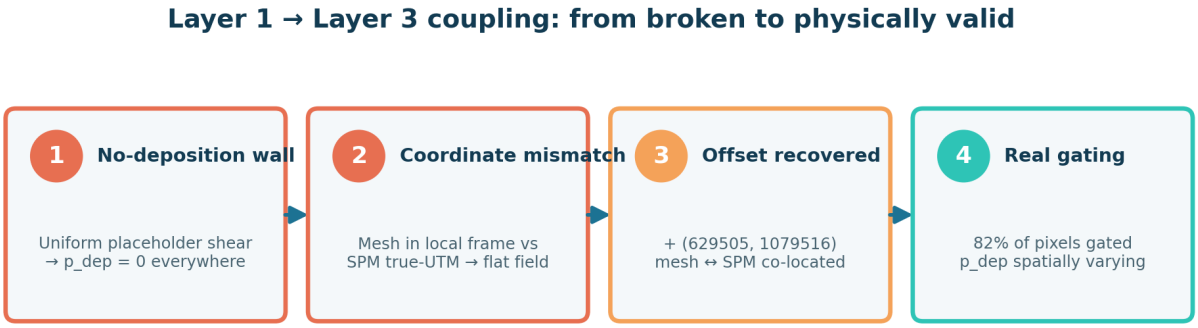


Figure 5.1: The coupling from broken to physically valid: the no-deposition wall and the coordinate mismatch, followed by offset recovery and real spatial gating.

5.2 The River-Forcing Investigation

An attempt to strengthen the hydrodynamic lever by adding Periyar discharge revealed that the mesh has only a seaward liquid boundary; there is no river inflow, and both candidate inlets lie on walled edges, as shown in Figure 5.2. The Periyar inlet, moreover, is a broad, deep basin rather than a narrow channel, so a properly-directed flow-rate boundary would carry very little velocity for a realistic discharge. A full solver-based river run was therefore deferred and the discharge folded into Sub-model A parametrically; the inlet nodes are preserved for the future solver run. A secondary lesson from this investigation was that the boundary-conditions file encodes the boundary rank in its line ordering (equivalently, in the mesh's IPOBO array), not in a trailing column that turned out to be a colour tag — a decoding error that had briefly corrupted the segment analysis.

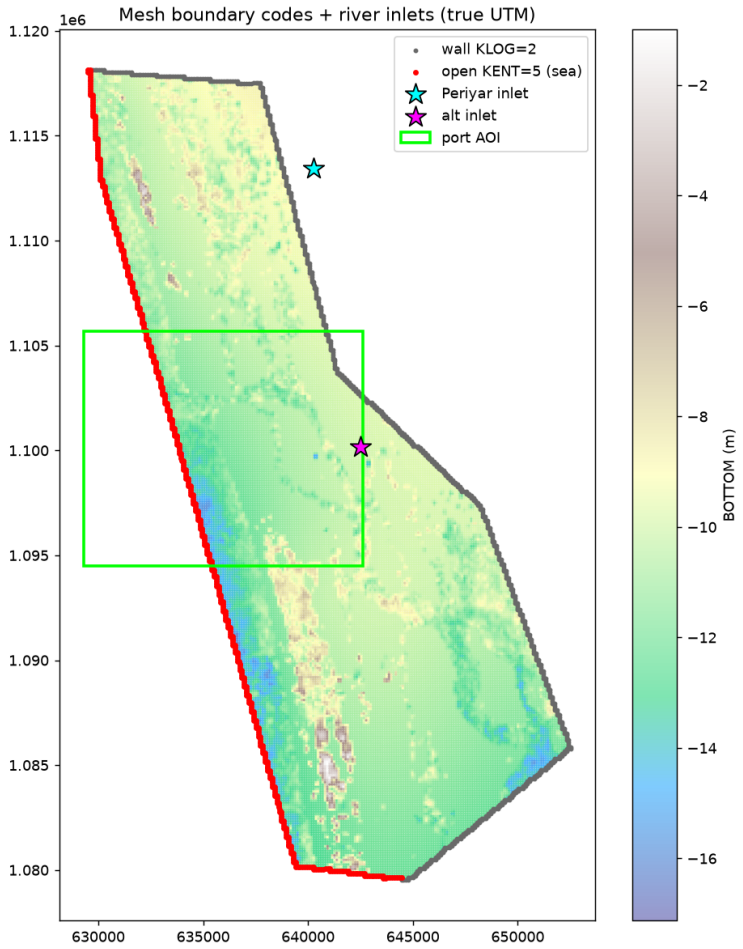


Figure 5.2: Mesh boundary in true UTM: the open sea boundary (red), walls (grey), the port area of interest (green), and the two candidate river inlets, both on walled edges.

5.3 Corrections to Sub-model A

The first end-to-end pipeline exhibited an implausible calibration (an effective near-bed enhancement below unity), a physically wrong zone split, and no comparison to the year-by-year record. Four corrections were applied and are summarised in Table 5.1.

#	Problem	Correction
C1	Volume integrated over the whole area of interest, overshooting the record by an order of magnitude	Integrate over the maintained-dredging footprint only; report the non-dredged estuary separately
C2	Raw signal set the zone split, over-weighting Mattancherry and under-weighting the basins	Impose the published fluid-mud structure as the zone weights; use satellite only for within-zone texture
C3	A single opaque scaling factor hid all error	Decompose into dredged-footprint fraction and an interpretable residual net-to-gross factor
C4	No inter-annual comparison; single-year illusion	Print the estimate against the multi-year dredging record and flag

Chapter 6

Prediction Validity

Because a HAINE output is only as reliable as its weakest upstream input, the pilot's outputs are placed on an explicit validity ladder, Figure 6.1. Four bands are distinguished: results that are independently correct; results that are correct only because they are pinned or imposed from a trusted source; results whose shape is right but whose magnitude is uncertain; and results that are structurally incomplete.

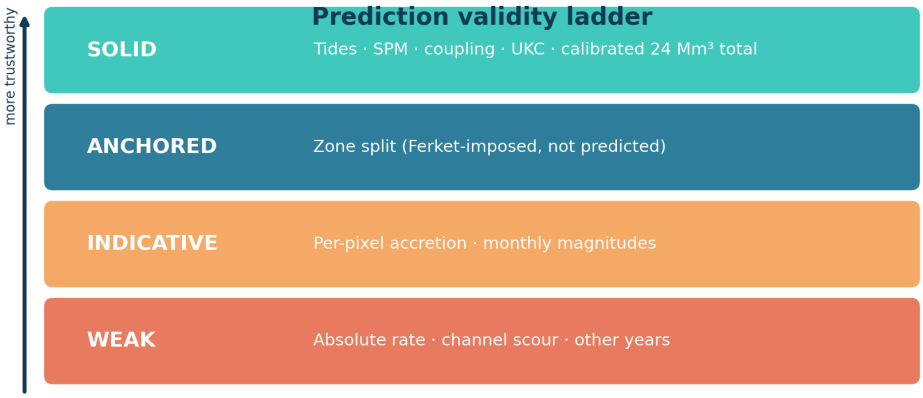


Figure 6.1: The prediction-validity ladder, from correct to structurally-incomplete outputs.

The tidal hydrodynamics, the satellite SPM product, the coupling geometry and the under-keel-clearance calculator are independently correct. The calibrated annual total and the per-zone split are correct but *anchored* — pinned to the dredging record and imposed from the published fluid-mud structure respectively, and therefore not independent predictions. The per-pixel accretion pattern and the monthly magnitudes are *indicative*: their relative structure is meaningful but their absolute values inherit the full proxy chain. The absolute deposition magnitude, the channel-scour component and any specific non-calibration year are *weak*, because the model is currently supply-dominated and label-free. Deviation concentrates physically in the outer channel, precisely where the missing wave and fluvial drivers matter most.

The data required to lift each output up the ladder is summarised in Figure 6.2. Two

datasets are decisive: paired pre- and post-dredge hydrographic surveys, which would supply per-location labels and convert the calibration into a trained regression, and multiple years of fused SPM, which would enable genuine inter-annual skill. Three further datasets — wave forcing, a solver-coupled river run, and measured Cochin sediment parameters — are accuracy upgrades rather than decisive gaps.

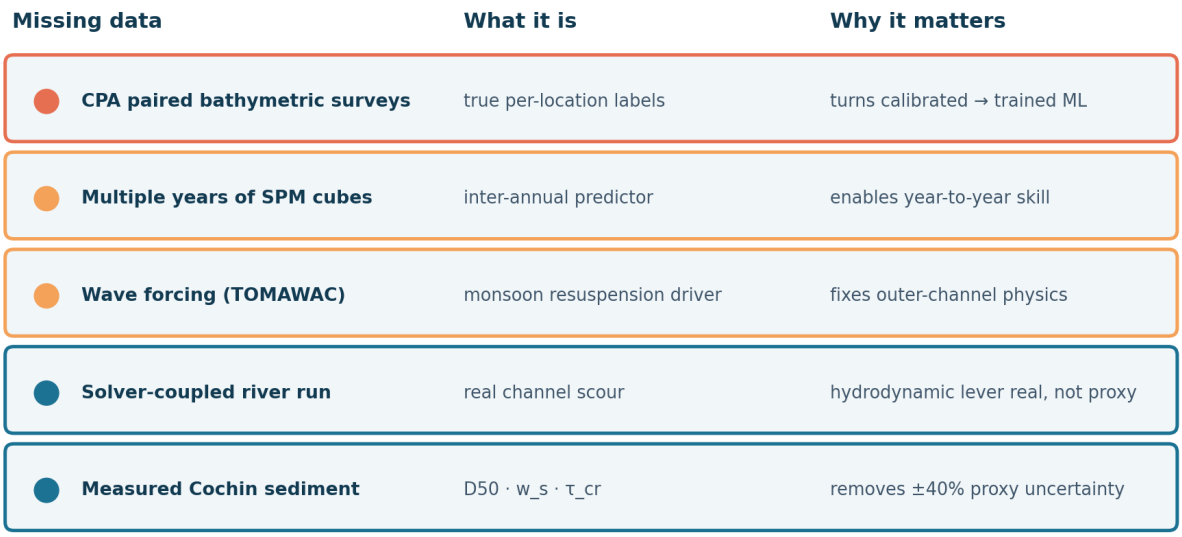


Figure 6.2: Missing data, what each is, and why it matters for lifting the pilot from a calibrated estimate towards a trained predictive model.

Chapter 7

Conclusions and Future Work

A working, end-to-end pilot of the HAINE framework has been implemented for Cochin Port. A validated tidal hydrodynamic model has been coupled, through a physically-correct deposition-gating field, to a satellite sediment product that independently reproduces the monsoon siltation cycle; from this coupling a calibrated siltation-volume assessment, an under-keel-clearance calculator and a decision dashboard have been produced. The pilot's honest status is that of a sound, literature-consistent decision-support baseline rather than a predictive siltation model: its total is calibrated and its zone split imposed, while its tidal hydrodynamics, satellite product and clearance calculations stand on their own.

The path from this baseline to a trained model is well defined and is shown in Figure 7.1. The immediate step is to complete the multi-year satellite extraction already initiated and to run an inter-annual calibration with leave-one-year-out cross-validation against the published dredging record, which will replace the single calibration factor with a validated constant and yield an honest error estimate. In parallel, the formal request to the Cochin Port Authority for paired survey data should be pursued, since these are the labels that would make the model trainable. The subsequent accuracy upgrades are the coupling of a wave model, a solver-based Periyar discharge run using the preserved inlet nodes, and the acquisition of measured sediment parameters. The framework is then ready to be extended to Paradip and Kandla, which represent contrasting siltation regimes, towards an Indian Ocean Region deployment.

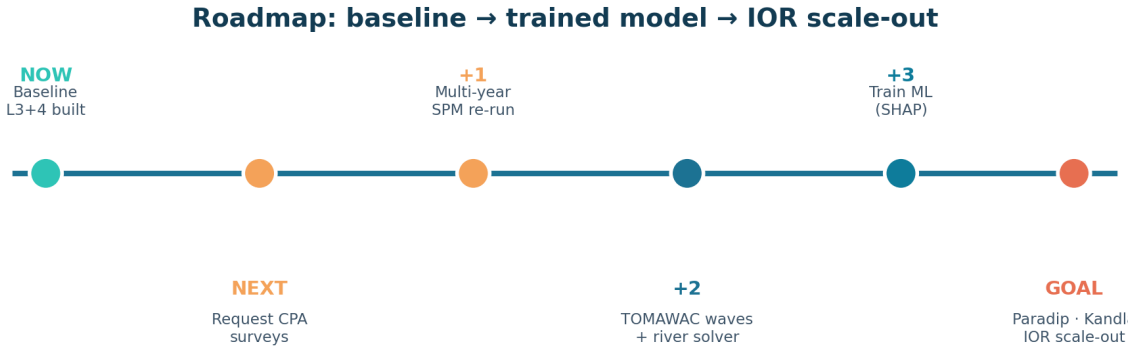


Figure 7.1: Roadmap from the delivered baseline, through multi-year calibration and physical upgrades, to a trained model and multi-port scale-out.

Chapter 8

Recommendations

1. **Prioritise the two decisive datasets.** Formally pursue paired pre/post-dredge hydrographic surveys through the Cochin Port Authority and the Sagar Samriddhi system, and complete the multi-year SPM extraction; these unblock a trained model where the accuracy upgrades only refine it.
2. **Couple a wave model to Layer 1.** Because Cochin siltation is substantially wave-driven and the reference study coupled waves explicitly, adding wave forcing (driven by ERA5) is the highest-value physical upgrade.
3. **Execute the solver-based river run** using the preserved Periyar inlet nodes, replacing the parametric discharge term with a real, directed flow that scours the channels.
4. **Acquire measured Cochin sediment parameters** (grain size, settling velocity and critical shear), since the sensitivity analysis shows these dominate the absolute-volume uncertainty.
5. **Structure all decision outputs to the MoPSW 2021 format** so that dredging work-orders and cost scenarios generated by Layer 4 are directly usable by port planners, and include the environmental-risk and explainability panels to build institutional confidence.
6. **Maintain the honest validity framing** in all reporting, distinguishing calibrated and imposed outputs from independent predictions, so that the tool is trusted for what it genuinely establishes.

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Appendices

Appendix A

Governing Equations

The cohesive erosion and deposition terms of the Partheniades–Krone framework are

$$E = M \left(\frac{\tau_b - \tau_{cr}}{\tau_{cr}} \right)^{n}, \qquad D = w_s \, C \left(1 - \frac{\tau_b}{\tau_{cd}} \right), \tag{A.1}$$

with M the erosion-rate coefficient, τ_{cr} and τ_{cd} the critical shear stresses for erosion and deposition, n the erosion exponent (unity for cohesive sediment), w_s the settling velocity and C the near-bed concentration. The tropical temperature correction to the critical shear stress is $\tau_{cr,T} = \tau_{cr,ref}\,[1 - 0.012\,(T - 15)]$, giving an approximately 18% reduction at a pre-monsoon sea-surface temperature of 30°C. The bed shear stress is estimated from the depth-averaged velocity as $\tau_b = \rho\, C_d\, |\mathbf{U}|^2$, and the Kochi effective-discharge term is $Q_{eff} = Q_{measured}(1 - \mathrm{DSR}) + \alpha_{coastal}\, C_{SPM,coast}$, where DSR is the upstream dam storage ratio.

Appendix B

Tools and Software Stack

Figure B.1 summarises every tool used across the four layers and the supporting workflow, together with its role.

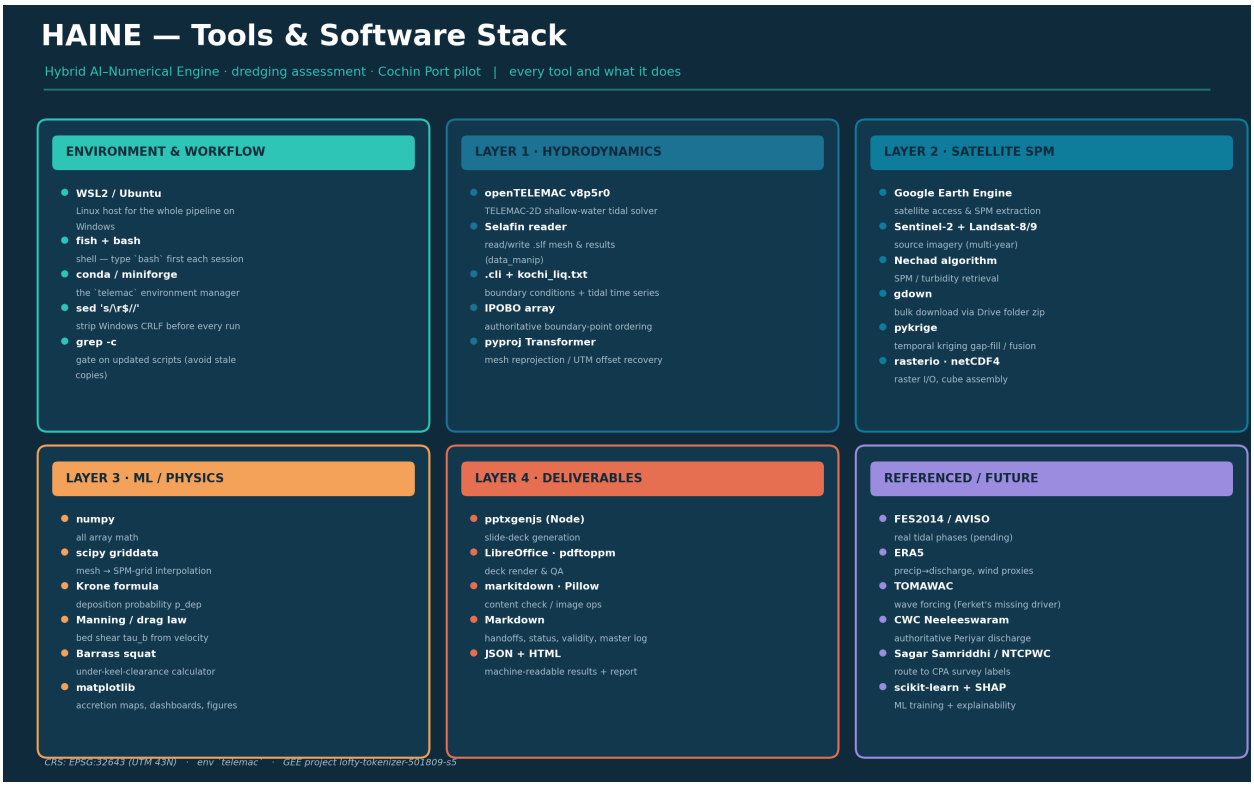


Figure B.1: The complete HAINE tools and software stack, organised by layer, with the use of each tool.

Appendix C

Key File and Script Inventory

Layer 1: the mesh and its degree-frame counterpart, the boundary-conditions and liquid-boundary files, the model control file, and the tidal result. Layer 2: the extraction, fusion and clipping scripts and the fused SPM cube. Layer 3: the bed-shear extraction, river-locating and segment scripts, the coupled pipeline and its revised version, and the multi-year extraction and calibration scripts. Layer 4: the decision dashboard, the machine-readable results file and the HTML report. A critical continuity note is that the solver mesh is stored in a local frame; a constant offset must be added to recover true UTM before any sampling of the hydrodynamic result against the satellite grid.